

The trajectory appears to cross itself repeatedly, but that's just an artifact of projecting the three-dimensional trajectory onto a two-dimensional plane. In three dimensions no self-intersections occur.

Let's try to understand Figure 9.3.2 in detail. The trajectory starts near the origin, then swings to the right, and then dives into the center of a spiral on the left. After a very slow spiral outward, the trajectory shoots back over to the right side, spirals around a few times, shoots over to the left, spirals around, and so on indefinitely. The number of circuits made on either side varies unpredictably from one cycle to the next. In fact, the sequence of the number of circuits has many of the characteristics of a *random* sequence. Physically, the switches between left and right correspond to the irregular reversals of the waterwheel that we observed in Section 9.1.

When the trajectory is viewed in all three dimensions, rather than in a two-dimensional projection, it appears to settle onto an exquisitely thin set that looks like a pair of butterfly wings. Figure 9.3.3 shows a schematic of this *strange attractor* (a term coined by Ruelle and Takens (1971)). This limiting set is the attracting set of zero volume whose existence was deduced in Section 9.2.

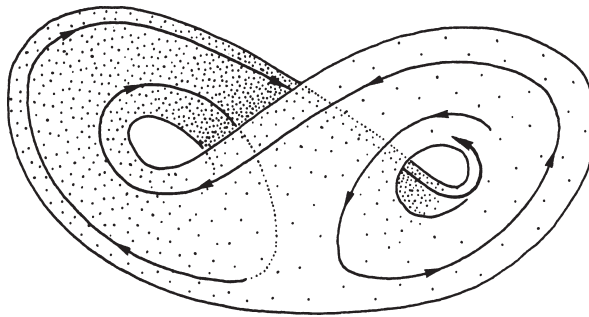


Figure 9.3.3 Abraham and Shaw (1983), p. 88

What is the geometrical structure of the strange attractor? Figure 9.3.3 suggests that it is a pair of surfaces that merge into one in the lower portion of Figure 9.3.3. But how can this be, when the uniqueness theorem (Section 6.2) tells us that trajectories can't cross or merge? Lorenz (1963) gives a lovely explanation—the two surfaces only *appear* to merge. The illusion is caused by the strong volume contraction of the flow, and insufficient numerical resolution. But watch where that idea leads him:

It would seem, then, that the two surfaces merely appear to merge, and remain distinct surfaces. Following these surfaces along a path parallel to a trajectory, and circling C^+ and C^- , we see that each surface is really a pair of surfaces, so that, where they appear to merge, there are really four surfaces. Continuing this process for another circuit, we see that there are

really eight surfaces, etc., and we finally conclude that there is an infinite complex of surfaces, each extremely close to one or the other of two merging surfaces.

Today this “infinite complex of surfaces” would be called a fractal. It is a set of points with zero volume but infinite surface area. In fact, numerical experiments suggest that it has a dimension of about 2.05! (See Example 11.5.1.) The amazing geometric properties of fractals and strange attractors will be discussed in detail in Chapters 11 and 12. But first we want to examine chaos a bit more closely.

Exponential Divergence of Nearby Trajectories

The motion on the attractor exhibits *sensitive dependence on initial conditions*. This means that two trajectories starting very close together will rapidly diverge from each other, and thereafter have totally different futures. Color Plate 2 vividly illustrates this divergence by plotting the evolution of a small red blob of 10,000 nearby initial conditions. The blob eventually spreads over the whole attractor. Hence nearby trajectories can end up anywhere on the attractor! The practical implication is that long-term prediction becomes impossible in a system like this, where small uncertainties are amplified enormously fast.

Let’s make these ideas more precise. Suppose that we let transients decay, so that a trajectory is “on” the attractor. Suppose $\mathbf{x}(t)$ is a point on the attractor at time t , and consider a nearby point, say $\mathbf{x}(t) + \delta(t)$, where δ is a tiny separation vector of initial length $\|\delta_0\| = 10^{-15}$, say (Figure 9.3.4).

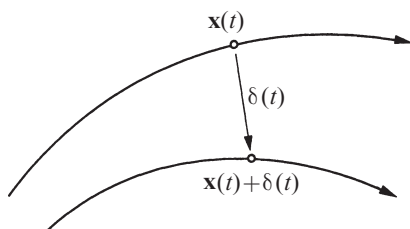


Figure 9.3.4

Now watch how $\delta(t)$ grows. In numerical studies of the Lorenz attractor, one finds that

$$\|\delta(t)\| \sim \|\delta_0\| e^{\lambda t}$$

where $\lambda \approx 0.9$. Hence *neighboring trajectories separate exponentially fast*. Equivalently, if we plot $\ln \|\delta(t)\|$ versus t , we find a curve that is close to a straight line with a positive slope of λ (Figure 9.3.5).

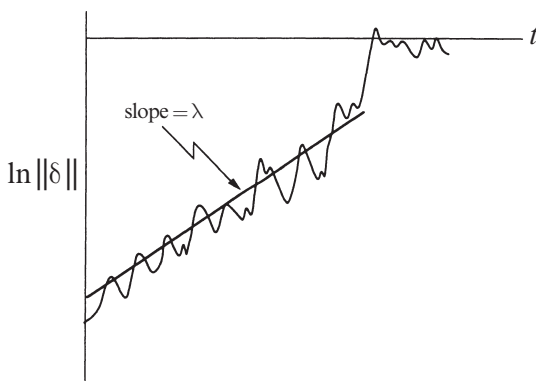


Figure 9.3.5

We need to add some qualifications:

1. The curve is never exactly straight. It has wiggles because the strength of the exponential divergence varies somewhat along the attractor.
2. The exponential divergence must stop when the separation is comparable to the “diameter” of the attractor—the trajectories obviously can’t get any farther apart than that. This explains the leveling off or *saturation* of the curve in Figure 9.3.5.
3. The number λ is often called the **Liapunov exponent**, although this is a sloppy use of the term, for two reasons:

First, there are actually n different **Liapunov exponents** for an n -dimensional system, defined as follows. Consider the evolution of an infinitesimal sphere of perturbed initial conditions. During its evolution, the sphere will become distorted into an infinitesimal ellipsoid. Let $\delta_k(t)$, $k = 1, \dots, n$, denote the length of the k th principal axis of the ellipsoid. Then $\delta_k(t) \sim \delta_k(0)e^{\lambda_k t}$, where the λ_k are the Liapunov exponents. For large t , the diameter of the ellipsoid is controlled by the most positive λ_k . Thus our λ is actually the *largest* Liapunov exponent.

Second, λ depends (slightly) on which trajectory we study. We should average over many different points on the same trajectory to get the true value of λ .

When a system has a positive Liapunov exponent, there is a *time horizon* beyond which prediction breaks down, as shown schematically in Figure 9.3.6. (See Lighthill 1986 for a nice discussion.) Suppose we measure the initial conditions of an experimental system very accurately. Of course, no measurement is perfect—there is always some error $\|\delta_0\|$ between our estimate and the true initial state.

from $+1$ and eventually reaches -1 . When this occurs, a flip bifurcation causes each of the black dots to split in two; the 3-cycle doubles its period and becomes a 6-cycle. The same mechanism operates here as in the original period-doubling cascade, but now produces orbits of period $3 \cdot 2^n$. A similar period-doubling cascade can be found in *all* of the periodic windows.

10.5 Liapunov Exponent

We have seen that the logistic map can exhibit aperiodic orbits for certain parameter values, but how do we know that this is really chaos? To be called “chaotic,” a system should also show *sensitive dependence on initial conditions*, in the sense that neighboring orbits separate exponentially fast, on average. In Section 9.3 we quantified sensitive dependence by defining the Liapunov exponent for a chaotic differential equation. Now we extend the definition to one-dimensional maps.

Here’s the intuition. Given some initial condition x_0 , consider a nearby point $x_0 + \delta_0$, where the initial separation δ_0 is extremely small. Let δ_n be the separation after n iterates. If $|\delta_n| \approx |\delta_0| e^{n\lambda}$, then λ is called the Liapunov exponent. A positive Liapunov exponent is a signature of chaos.

A more precise and computationally useful formula for λ can be derived. By taking logarithms and noting that $\delta_n = f^n(x_0 + \delta_0) - f^n(x_0)$, we obtain

$$\begin{aligned}\lambda &\approx \frac{1}{n} \ln \left| \frac{\delta_n}{\delta_0} \right| \\ &= \frac{1}{n} \ln \left| \frac{f^n(x_0 + \delta_0) - f^n(x_0)}{\delta_0} \right| \\ &= \frac{1}{n} \ln |(f^n)'(x_0)|\end{aligned}$$

where we’ve taken the limit $\delta_0 \rightarrow 0$ in the last step. The term inside the logarithm can be expanded by the chain rule:

$$(f^n)'(x_0) = \prod_{i=0}^{n-1} f'(x_i) .$$

(We’ve already seen this formula in Example 9.4.1, where it was derived by heuristic reasoning about multipliers, and in Example 10.3.3, for the special case $n = 2$.) Hence

$$\begin{aligned}\lambda &\approx \frac{1}{n} \ln \left| \prod_{i=0}^{n-1} f'(x_i) \right| \\ &= \frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)|.\end{aligned}$$

If this expression has a limit as $n \rightarrow \infty$, we define that limit to be the *Liapunov exponent* for the orbit starting at x_0 :

$$\lambda = \lim_{n \rightarrow \infty} \left\{ \frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)| \right\}.$$

Note that λ depends on x_0 . However, it is the same for all x_0 in the basin of attraction of a given attractor. For stable fixed points and cycles, λ is negative; for chaotic attractors, λ is positive.

The next two examples deal with special cases where λ can be found analytically.

EXAMPLE 10.5.1:

Suppose that f has a stable p -cycle containing the point x_0 . Show that the Liapunov exponent $\lambda < 0$. If the cycle is superstable, show that $\lambda = -\infty$.

Solution: As usual, we convert questions about p -cycles of f into questions about fixed points of f^p . Since x_0 is an element of a p -cycle, x_0 is a fixed point of f^p . By assumption, the cycle is stable; hence the multiplier $|(f^p)'(x_0)| < 1$. Therefore $\ln |(f^p)'(x_0)| < \ln(1) = 0$, a result that we'll use in a moment.

Next observe that for a p -cycle,

$$\begin{aligned} \lambda &= \lim_{n \rightarrow \infty} \left\{ \frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)| \right\} \\ &= \frac{1}{p} \sum_{i=0}^{p-1} \ln |f'(x_i)| \end{aligned}$$

since the same p terms keep appearing in the infinite sum. Finally, using the chain rule in reverse, we obtain

$$\frac{1}{p} \sum_{i=0}^{p-1} \ln |f'(x_i)| = \frac{1}{p} \ln |(f^p)'(x_0)| < 0,$$

as desired. If the cycle is superstable, then $|(f^p)'(x_0)| = 0$ by definition, and thus $\lambda = \frac{1}{p} \ln(0) = -\infty$. ■

The second example concerns the *tent map*, defined by

$$f(x) = \begin{cases} rx, & 0 \leq x \leq \frac{1}{2} \\ r - rx, & \frac{1}{2} \leq x \leq 1 \end{cases}$$

for $0 \leq r \leq 2$ and $0 \leq x \leq 1$ (Figure 10.5.1).

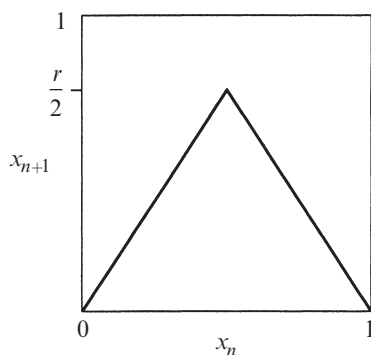


Figure 10.5.1

Because it is piecewise linear, the tent map is far easier to analyze than the logistic map.

EXAMPLE 10.5.2:

Show that $\lambda = \ln r$ for the tent map, independent of the initial condition x_0 .

Solution: Since $f'(x) = \pm r$ for all x , we find $\lambda = \lim_{n \rightarrow \infty} \left\{ \frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)| \right\} = \ln r. \blacksquare$

Example 10.5.2 suggests that the tent map has chaotic solutions for all $r > 1$, since $\lambda = \ln r > 0$. In fact, the dynamics of the tent map can be understood in detail, even in the chaotic regime; see Devaney (1989).

In general, one needs to use a computer to calculate Liapunov exponents. The next example outlines such a calculation for the logistic map.

EXAMPLE 10.5.3:

Describe a numerical scheme to compute λ for the logistic map $f(x) = rx(1 - x)$. Graph the results as a function of the control parameter r , for $3 \leq r \leq 4$.

Solution: Fix some value of r . Then, starting from a random initial condition, iterate the map long enough to allow transients to decay, say 300 iterates or so. Next compute a large number of additional iterates, say 10,000. You only need to store the current value of x_n , not all the previous iterates. Compute $\ln |f'(x_n)| = \ln |r - 2rx_n|$ and add it to the sum of the previous logarithms. The Liapunov exponent is then obtained by dividing the grand total by 10,000. Repeat this procedure for the next r , and so on. The end result should look like Figure 10.5.2.

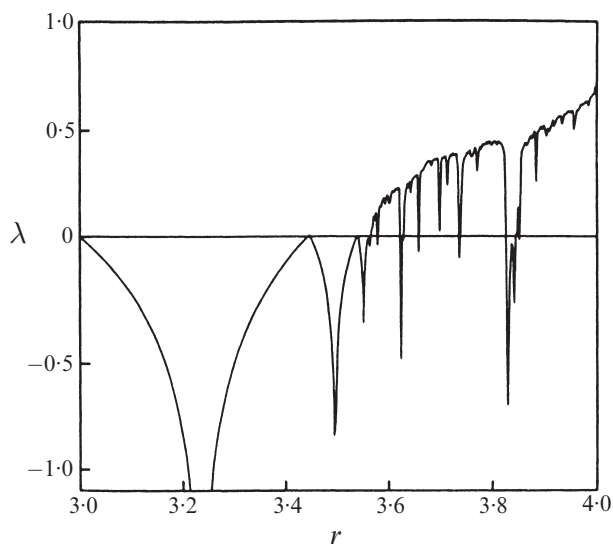


Figure 10.5.2 Olsen and Degn (1985), p. 175

Comparing this graph to the orbit diagram (Figure 10.2.7), we notice that λ remains negative for $r < r_\infty \approx 3.57$, and approaches zero at the period-doubling bifurcations. The negative spikes correspond to the 2^n -cycles. The onset of chaos is visible near $r \approx 3.57$, where λ first becomes positive. For $r > 3.57$ the Liapunov exponent generally increases, except for the dips caused by the windows of periodic behavior. Note the large dip due to the period-3 window near $r = 3.83$. ■

Actually, all the dips in Figure 10.5.2 should drop down to $\lambda = -\infty$, because a superstable cycle is guaranteed to occur somewhere near the middle of each dip, and such cycles have $\lambda = -\infty$, by Example 10.5.1. This part of the spike is too narrow to be resolved in Figure 10.5.2.

10.6 Universality and Experiments

This section deals with some of the most astonishing results in all of nonlinear dynamics. The ideas are best introduced by way of an example.

EXAMPLE 10.6.1:

Plot the graph of the **sine map** $x_{n+1} = r \sin \pi x_n$ for $0 \leq r \leq 1$ and $0 \leq x \leq 1$, and compare it to the logistic map. Then plot the orbit diagrams for both maps, and list some similarities and differences.

Solution: The graph of the sine map is shown in Figure 10.6.1.